

Summary

Research Interests: Numerical linear algebra; communication-avoiding and parallel algorithms; large-scale and stochastic optimization; sparse matrix and tensor computations; high performance computing for machine learning.

Teaching Interests:

Core Courses: Undergraduate/Graduate-level Algorithms.

Elective Courses: Parallel Algorithms, Numerical Optimization, Scientific AI and Machine Learning.

Publications: 3 refereed journal articles, 11 refereed conference papers (SC, NeurIPS, IPDPS, VLDB, SIAM PP, HPCAsia), 5 preprints/manuscripts under review

Citations: 775; **h-index:** 8 (Google Scholar, July 2026)

Recognition: Outstanding Paper Award, HPCAsia 2025

Grants Awarded: 5 awards (2 DOE ASCR, Co-PI; 1 ORNL, PI; 1 NSF student travel, PI; 1 internal, PI) and 2 leadership-computing allocations (ALCC, NERSC ERCAP)

Students Advised: 11 graduate and 11 undergraduate students; 1 undergraduate honors thesis

Web: aditya08.github.io

GitHub/GitLab: [aditya08](#)

ORCID: [0000-0002-8251-9150](#)

Professional Appointments

Assistant Professor 2022 – current
Department of Computer Science, Wake Forest University

Assistant Research Scientist (Postdoctoral Position) 2019 – 2022
Institute for Data Intensive Engineering and Science, Johns Hopkins University

Industry and Lab Experience

Oak Ridge National Laboratory 2024 – 2025
Visiting Scientist

NVIDIA Research 2017
Research Internship

Education

Ph.D. in Computer Science, University of California, Berkeley 2018
Designated Emphasis in Computational and Data Science and Engineering
Thesis: *Avoiding Communication in First Order Methods for Optimization*
Advised by Jim Demmel and Michael W. Mahoney

M.S. in Computer Science, University of California, Berkeley 2016

B.S. in Electrical and Computer Engineering, Rutgers University 2012
Double major in Computer Science
Honors Thesis: *Scaling Replica Exchange Molecular Dynamics*
Advised by Manish Parashar and Ivan Roderio

Grants and Awarded Computing Allocations

Wake Forest University

PI, Wake Forest University: Leadership & Character Course Development Grant	2026
PI, Oak Ridge National Laboratory: Subcontract	2024 – 2025
Co-PI, DOE ASCR: Data Reduction for Science	2024 – 2027
PI, NSF OAC: Travel: Student Travel Grant for ICPP 2023	2023 – 2024
Co-PI, DOE ASCR: Mathematical Multifaceted Integrated Capability Center (MMICC) <i>Sparsitute: A Mathematical Institute for Sparse Computations in Science and Engineering</i>	2022 – 2027

Compute Time

Co-PI, An Exascale Co-Design Framework for Spiking Graph Neural Networks <i>ASCR ALCC</i>	2025 – current
PI, Scalable Numerical Algorithms and Libraries for Sparse Computations <i>NERSC ERCAP</i>	2024 – current

Honors and Awards

HPC-Asia 2025, Outstanding Paper Award	2025
NSF Graduate Research Fellowship	2013 – 2018
UC Berkeley EECS Department Fellowship	2012 – 2013
Rutgers School of Engineering James J. Slade Scholar	2012
James L. Potter Award in Research	2012
Edward J. Bloustein Distinguished Scholars Award	2008 – 2012
Rutgers Scarlet Scholarship	2008 – 2012
Kuhl Memorial Engineering Fellowship	2011
Charles V. Longo Memorial Scholarship	2011
Rutgers Academic Excellence Award	2010
Hannah Sands Endowed Scholarship in Engineering	2010

Publications

Refereed Journal Articles

1. **Aditya Devarakonda**, Kimon Fountoulakis, James Demmel, and Michael W Mahoney. Avoiding communication in primal and dual block coordinate descent methods. *SIAM Journal on Scientific Computing*, 41(1):C1–C27, 2019.
2. Manish Parashar, Moustafa AbdelBaky, Ivan Roderio, and **Aditya Devarakonda**. Cloud paradigms and practices for computational and data-enabled science and engineering. *Computing in Science & Engineering*, 15(4):10–18, 2013.
3. David Villegas, Norman Bobroff, Ivan Roderio, Javier Delgado, Yanbin Liu, **Aditya Devarakonda**, Liana Fong, S Masoud Sadjadi, and Manish Parashar. Cloud federation in a layered service model. *Journal of Computer and System Sciences*, 78(5):1330–1344, 2012.

Refereed Conference and Workshop Papers

1. Julian Bellavita, Matthew Rubino, Nakul Iyer, Andrew Chang, **Aditya Devarakonda**, Flavio Vella, and Giulia Guidi. Communication-Avoiding Linear Algebraic Kernel K-Means on GPUs. In *Proceedings of the 40th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*. IEEE, 2026.
2. Selahattin Akkas, **Aditya Devarakonda**, and Ariful Azad. DistShap: Scalable GNN explanations with distributed shapley values. *arXiv:2506.22668*, 2025. **Accepted, VLDB'26**.
3. Joao Pinheiro, **Aditya Devarakonda**, and Grey Ballard. Parallel rank-adaptive higher order orthogonal iteration. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, SC '25*, page 1800–1815, New York, NY, USA, 2025. Association for Computing Machinery.
4. Zishan Shao and **Aditya Devarakonda**. Scalable dual coordinate descent for kernel methods. In *Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region, HPCASIA '25*, page 52–63, New York, NY, USA, 2025. Association for Computing Machinery. **Outstanding Paper Award**.
5. Yixiao Wang, Zishan Shao, Ting Jiang, and **Aditya Devarakonda**. Enhanced cyclic coordinate descent methods for elastic net penalized linear models. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. Curran Associates, Inc., 2025.
6. **Aditya Devarakonda** and Grey Ballard. Sequential and shared-memory parallel algorithms for partitioned local depths. In *Proceedings of the 2024 SIAM Conference on Parallel Processing for Scientific Computing (PP)*, pages 53–64. SIAM, 2024.
7. **Aditya Devarakonda** and James Demmel. Avoiding communication in logistic regression. In *2020 IEEE 27th International Conference on High Performance Computing, Data, and Analytics (HiPC)*, pages 91–100, Los Alamitos, CA, USA, 2020. IEEE Computer Society.
8. **Aditya Devarakonda**, Kimon Fountoulakis, James Demmel, and Michael W Mahoney. Avoiding synchronization in first-order methods for sparse convex optimization. In *2018 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 409–418. IEEE, 2018.
9. Saeed Soori, **Aditya Devarakonda**, Zachary Blanco, James Demmel, Mert Gurbuzbalaban, and Maryam Mehri Dehnavi. Reducing communication in proximal newton methods for sparse least squares problems. In *Proceedings of the 47th International Conference on Parallel Processing, ICPP 2018*, pages 22:1–22:10, New York, NY, USA, 2018. ACM.
10. Alex Gittens, **Aditya Devarakonda**, Evan Racah, Michael Ringenburt, Lisa Gerhardt, Jey Kottalam, Jialin Liu, Kristyn Maschhoff, Shane Canon, Jatin Chhugani, et al. Matrix factorizations at scale: A comparison of scientific data analytics in spark and c+ mpi using three case studies. In *2016 IEEE International Conference on Big Data (Big Data)*, pages 204–213. IEEE, 2016.
11. Razvan Carbunescu, **Aditya Devarakonda**, James Demmel, Steven Gordon, Jay Alameda, and Susan Mehringer. Architecting an autograder for parallel code. In *Proceedings of the 2014 Annual Conference on Extreme Science and Engineering Discovery Environment, XSEDE '14*, New York, NY, USA, 2014. Association for Computing Machinery.

Preprints, Manuscripts Under Review, and Technical Reports

1. **Aditya Devarakonda**, Irene Simó Muñoz, and Giulia Guidi. Mixed-precision communication-avoiding SGD for generalized linear models on GPUs. *arXiv:2606.18463*, 2026. **Under review**, ACM Trans. Math. Softw.
2. **Aditya Devarakonda** and Ramakrishnan Kannan. Communication-efficient, 2D parallel stochastic gradient descent for distributed-memory optimization. *arXiv:2501.07526*, 2025. **Under review**.
3. Ramkrishnan Kannan, **Aditya Devarakonda**, Prasanna Date, Ashish Gautam, Shruti Kulkarni, Hao Lu, Jakub Kurzak, Guojing Cong, Seung-Hwan Lim, Catherine Schuman, Robert Patton, Gina Adam, Yiran Chen, and Thomas Potok. HyperNeuro: A high performance simulator for neuromorphic computing. 2025. **Under review**.
4. **Aditya Devarakonda**, Maxim Naumov, and Michael Garland. Adabatch: adaptive batch sizes for training deep neural networks. *arXiv preprint arXiv:1712.02029*, 2017.
5. Nishant Totla and **Aditya Devarakonda**. Massive parallelization of sat solvers. Technical report, University of California, Berkeley, 2013.

Research Software

Mixed Precision Communication-Avoiding SGD — Mixed-precision CA-SGD for generalized linear models on multi-GPU systems (BSD 3-Clause).

doi.org/10.5281/zenodo.20448273

Lead developer

Vivaldi — Distributed-memory 1D/1.5D kernel K -means clustering for multi-GPU systems.

github.com/CornellHPC/Vivaldi

Contributor

HybridSGD — 2D parallel, communication-efficient stochastic gradient descent (C/MPI).

github.com/aditya08/hybridsgd

Lead developer

TuckerMPI-HOOI — Distributed rank-adaptive Higher-Order Orthogonal Iteration for Tucker decomposition.

doi.org/10.5281/zenodo.16752647

Co-developer

DistSHAP — Distributed Shapley-value explanations for graph neural networks.

doi.org/10.5281/zenodo.15272858

Co-developer

AdaBatch — Adaptive batch-size training for deep neural networks (NVIDIA Labs).

github.com/NVLabs/AdaBatch

Lead developer

Talks and Presentations

Invited Talks, Seminars, and Colloquia

1. Physics-informed neural networks and scalable optimization algorithms. In *Department of Physics Seminar, Wake Forest University*, November 2023.
2. Scalable algorithms and techniques for machine learning. In *Graduate Thesis Seminar, Wake Forest University*, October 2023.
3. Hybrid 2D s -step methods for machine learning. In *Mathematics in Computation Seminar, Oak Ridge National Laboratory*, July 2023.
4. s -step methods for machine learning and beyond. In *Sparsitute Bi-Weekly Seminar*, February 2023.
5. Scalable algorithms and techniques for machine learning. In *Graduate Thesis Seminar, Wake Forest University*, October 2022.
6. Avoiding communication in first-order methods for optimization. In *Invited talk, Reservoir Labs*, January 2021.
7. Avoiding communication in logistic regression. In *CS Seminar, Lawrence Berkeley National Laboratory*, January 2019.
8. Avoiding synchronization in first-order methods for convex optimization. In *Dissertation Talk*, May 2018.
9. Avoiding synchronization in first-order methods for sparse convex optimization. In *UC Berkeley Scientific Computing and Matrix Computations Seminar*, February 2018.
10. Avoiding communication in machine learning. In *Johns Hopkins University Applied Physics Lab*, January 2018.
11. Communication-avoiding algorithms. In *NASA Langley Research Center*, December 2017.
12. Communication-avoiding krylov subspace methods. In *NASA Langley Research Center*, December 2017.
13. Adabatch: Adaptive batch sizes for training deep neural networks. In *NVIDIA*, November 2017.
14. Communication-avoiding coordinate descent methods for linear systems. In *UC Berkeley Scientific Computing and Matrix Computations Seminar*, March 2016.

Conference and Workshop Presentations

1. Hybrid s-step sgd and local sgd for large scale optimization. In *SIAM Conference on Mathematics of Data Science (MDS24)*, October 2024.
2. Scalable dual coordinate descent for kernel methods. In *SIAM Conference on Applied Linear Algebra (LA24)*, May 2024.
3. Sequential and shared-memory parallel algorithms for partitioned local depths. In *SIAM Conference on Parallel Processing (PP24)*, March 2024.
4. s-step dual coordinate descent methods for kernelized classification and regression. In *DOE ASCR PI Meeting*, January 2024.
5. Hybrid 2D s-step methods for machine learning. In *All-Members Meeting, Sparsitute*, October 2023.
6. 2D s-step methods for machine learning. In *SIAM Conference on Computational Science and Engineering (CSE23)*, February 2023.
7. Avoiding communication in logistic regression. In *International Conference on High Performance Computing, Data, and Analytics*, December 2020.
8. Avoiding synchronization in first-order methods for sparse convex optimization. In *International Parallel and Distributed Processing Symposium*, May 2018.
9. s-step methods in machine learning. In *SIAM Parallel Processing*, March 2018.
10. Communication-avoiding machine learning. In *Householder Symposium XX*, July 2017.
11. Communication-avoiding methods for regularized least-squares. In *SIAM Annual*, July 2017.
12. Communication-avoiding machine learning. In *Householder Symposium XX*, June 2017.
13. Communication avoiding primal and dual block coordinate descent methods. In *SIAM Optimization*, May 2017.
14. A generalized framework for communication-avoiding regularized least squares. In *Berkeley Statistics Annual Research Symposium*, April 2017.
15. Matrix factorizations at scale: A comparison of scientific data analytics in spark and c+ mpi using three case studies. In *IEEE Big Data*, July 2016.
16. Avoiding communication in machine learning. In *Berkeley Statistics Annual Research Symposium*, April 2016.
17. Scalable asynchronous molecular dynamics on federated cyberinfrastructure. In *SCALE Challenge (CCGRID)*, May 2012.
18. Parallel quicksort. In *Passionate on Parallel Research Experience for Undergraduates*, August 2011.
19. Molecular dynamics on federated cyberinfrastructure. In *Life Sciences and Pharmaceuticals Industrial Advisory Board*, May 2011.

Teaching

Wake Forest University

CSC 301: Algorithms Design and Analysis	Fa23, Sp24
CSC 721: Theory of Algorithms	Sp24, Sp26
CSC 393: Individual Study: <i>s-step and communication-avoiding algorithms</i>	Fa23
CSC 790: Parallel Numerical Optimization	Sp23
CSC 393: Individual Study: <i>sparse matrix computations</i>	Sp23
CSC 201: Data Structures and Algorithms	Sp22 – Fa23

Johns Hopkins University

AS117.205: Beautiful Data Sp19, Sp20
Guest Lecturer

Johns Hopkins University Center for Talented Youth

Introduction to Deep Learning 01/2020
Workshop Presenter

Introduction to Electrical Engineering Sum10, Sum14
Teaching Assistant

University of California, Berkeley

CS267: Applications of Parallel Computers Sp14, Sum16, Sp18
Graduate Student Instructor

CS267: Applications of Parallel Computers Sp17
Guest Lecturer

ParLab Parallel Boot Camp: Short Course on Parallel Computing Sum14
Teaching Assistant

Rutgers University, New Brunswick

CS111: Introduction to Computer Science Sp10
Teaching Assistant

Advising and Mentorship

Graduate Students — Current

Golam Mahmud, M.S., Wake Forest University

Graduate Students — Former Wake Forest Advisees

Zishan Shao, M.S. *now: Ph.D. student, Duke University*

João Pinheiro, M.S. (co-advised with Grey Ballard) *now: Ph.D. student, Purdue University*

Xiaotian Liu, M.S. *now: Ph.D. student, Dartmouth College*

Qingjia Xu, M.S., Wake Forest University *now: Oracle*

Ziyue Yang, M.S. *now: Intel, Shanghai*

Nikhil Rajkumar, M.S. *now: Red Ventures*

Alex Michon, M.S. *now: Arista Networks*

Graduate Students — External Research Mentees

Irene Simó Muñoz, Ph.D. student, Cornell University

Alan de Mersseman, M.Eng. student, Cornell University

Yixiao Wang, M.S. student, Duke University *now: Ph.D. student, Duke University*

Advisee Conference Presentations

- Enhanced Cyclic Coordinate Descent Methods for Elastic Net Penalized Linear Models* December 2025
Poster presented by student co-author, Conference on Neural Information Processing Systems (NeurIPS 2025)
- Scalable Dual Coordinate Descent for Kernel Methods* January 2025
Presented by Zishan Shao, International Conference on High Performance Computing in Asia-Pacific Region (HP-CAAsia 2025). **Outstanding Paper Award**

Undergraduate Research Students

- AJ Garcia
- Ting (Justin) Jiang *now: Ph.D. student, CMU*
- Boyuan Pan *M. S. student, University of Pennsylvania*
- Mason Horne *now: Intercontinental Exchange*
- Richard Zhang *now: M.S. student, U. Pennsylvania*
- Ziang Xu *now: M.S. student, Columbia University*
- Rachel Lin *now: Ph.D. student, Indiana University*
- Runhao Huang *now: Morgan Stanley*
- Cecilia Liu *now: Microsoft*
- Jieming Wei *now: Uber*
- Jiachen Liu *now: AppDev*

Service

Editorial and Organizational Leadership

- Frontiers in High Performance Computing* Review Editor, 2023 – current
- IEEE Technical Community on High Performance Computing Co-Chair, Education and Outreach Initiative
- International Conference on Parallel Processing (ICPP 2023) Poster & Student Program Co-Chair
- SC22/SC23 Birds of a Feather: TCHPC Career Panel Co-Organizer and Panel Moderator
- TAPIA 2024: *There's no GPT for this: Graduate School and What Lies Beyond* Invited Panelist

Invited Teaching and Outreach

- Cornell University CS 5220: Applied High-Performance and Parallel Computing Guest Lecturer, Sp26

Wake Forest University

- Department of Computer Science Tenure-Track Hiring Committee Member, 2024
- Department of Computer Science Undergraduate Academic Advisor 2024 – current
- Department of Computer Science Curriculum Committee Member, 2024 – current
- Department of Computer Science Graduate Committee Member, 2022 – 2023

University of California, Berkeley

EECS Peers Computer Science Coordinator	2016 – 2018
UC Berkeley BiasBusters Workshop Presenter	2017 – 2018
UC Berkeley Graduate Assembly EECS Delegate	2016 – 2017
Information Risk Governance Committee Member	2016 – 2017
Campus Information Security and Privacy Committee Member	2016 – 2017
UC Berkeley EECS Faculty Search Committee Student Member	2015
UC Berkeley EECS Graduate Admissions Committee Student Member	2015

External Reviewing

Conferences and Workshops

IA³ 2026: Workshop on Irregular Applications: Architectures and Algorithms (at SC26)

IEEE International Conference on Big Data (BigData 2026)

ACM International Conference on Parallel Processing (ICPP 2026)
Algorithms Track

ACM/IEEE SC 2026
Algorithms Track

ACM Conference on Web Search and Data Mining (WSDM 2026)
Demo Track

SIAM Conference on Parallel Processing for Scientific Computing (PP 2026)

ISC High Performance 2025, 2026
Machine Learning and AI Track

IPDPSW GRAPL 2025: Workshop on Graphs, Architectures, Programming, and Learning

International Conference on High Performance Computing in Asia-Pacific Region (HPCAsia 2025)
Machine Learning, AI, and Emerging Technologies Track

ACM/IEEE International Conference on Utility and Cloud Computing (UCC 2024)

IPDPSW GRAPL 2024: Workshop on Graphs, Architectures, Programming, and Learning

SIAM Conference on Parallel Processing for Scientific Computing (PP 2024)

IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC) 2024
Scalable Algorithms and Analytics Track

International Conference on High Performance Computing in Asia-Pacific Region (HPCAsia 2024)
Application and Algorithm Track

ACM/IEEE International Conference on Utility and Cloud Computing (UCC 2023)

IPDPS PhD Forum 2023

ACM/IEEE SC 2021, SC 2022, and SC 2023
Machine Learning and HPC Track

ACM/IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2022)
AI, ML, and Deep Learning Track

IEEE International Conference on Cluster Computing Cluster 2022
Applications, Algorithms, and Libraries Track

Journals

- SIAM Journal on Matrix Analysis and Applications (SIMAX)
- Journal of Parallel and Distributed Computing (JPDC)
- ACM Transactions on Mathematical Software (TOMS)
- ACM Transactions on Parallel Computing (TOPC)
- IEEE Transactions on Artificial Intelligence (TAI)
- Journal of Astronomy and Computing

Proposals

- | | |
|---|------------------|
| Department of Defense (NDSEG Fellowship Reviewer) | 2025 |
| National Science Foundation | 2024, 2026 |
| Department of Energy | 2023, 2025, 2026 |

Professional Memberships

- ACM, IEEE, and SIAM (SIAG/Supercomputing, SIAG/Linear Algebra)